

Total No. of Questions : 8]

SEAT No. :

P812

[5870] - 1133

[Total No. of Pages : 2

**T.E. (Computer Engineering)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - II) (310251)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 1) Neat diagrams must be drawn whenever necessary.
- 2) Figures to the right side indicate full marks.
- 3) Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam tables is allowed.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is driving data deluge? Explain with one example. [9]
- b) What is data science? Differentiate between Business Intelligence and Data Science. [9]

OR

- Q2)** a) What are the sources of Big Data. Explain model building phase with example. [9]
- b) Explain big data analytics architecture with diagram. What is data discovery phase. Explain with example. [9]

- Q3)** a) Explain various data pre-processing steps. Discuss essential python libraries for preprocessing. [8]
- b) What are association rules? Explain Apriori Algorithm in brief. [9]

OR

- Q4)** a) Explain the following
- i) Linear Regression
 - ii) Logistic Regression [8]
- b) Explain scikit-learn library for matplotlib with example. [9]

P.T.O.

- Q5) a)** Write short note on
- i) Time series Analysis
 - ii) TF - IDF. [9]
- b) What is clustering? With suitable example explain the steps involved in k - means algorithm. [9]

OR

- Q6) a)** Write short note on
- i) Confusion matrix
 - ii) AUC - ROC curve [9]
- b) Discuss Holdout method and Random Sub Sampling methods. [9]
- Q7) a)** With a suitable example explain Histogram and explain its usages. [8]
- b) Describe the Data visualization tool “Tableau”. Explain its applications in brief. [9]

OR

- Q8) a)** With a suitable example explain and draw a Box plot and explain its usages. [8]
- b) Describe the challenges of data visualization. Draw box plot and explain its usages. [9]

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Total No. of Questions : 8]

SEAT No. :

PA-1449

[Total No. of Pages : 3

[5926]-65

T.E. (Computer Engg.)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester-II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3, or Q4, Q5 or Q6, and Q7 or Q8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks,
- 4) Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam tables is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Draw the diagram of data analytics life cycle in big data and briefly explain its phases. [8]

b) Explain in detail how the model building phase is built by team in data analytics life cycle? [9]

OR

Q2) a) List and explain the steps in data preparation phase of data analytics life cycle. [8]

b) Write short note on the following: [9]

- i) ETL
- ii) Common tools for the model building.
- iii) Model selection for data analytics.

Q3) a) What are the types of analytics in big data? Explain in brief. [9]

b) Calculate the support and confidence value for all the possible item sets. [9]

Transaction ID	Items bought
1	Onion, Potato, Cold drink
2	Onion, Burger, Cold drink
3	Eggs, Onion, Cold drink
4	Potato, Milk, Eggs.
5	Potato, Burger, cold drink, Milk eggs.

OR

P.T.O.

- Q4)** a) Explain the use of logistic function in logistic regression in detail. [9]
 b) Write short note on the following:
 i) Removing duplicates from data set.
 ii) Handling missing data
 iii) Data transformation [9]

- Q5)** a) Suppose that the given data the taste is to cluster points (With (x,y) representing location) into three cluster, where the points are.
 A1(2,10), A2(2,5), A3(8,4), B1 (5,8)
 B2(7,5) B3(6,4), C1(1,2), C2(4,9)
 The distance function is Euclidean distance suppose initially we assign A1, B1 and C1 as the center of each cluster, respectively. use the k-means algorithm to show only the three cluster centers after the first round of execution with steps. [9]
 b) Explain the following text analysis steps with suitable example. [8]
 i) Part of speech (POS) tagging
 ii) Lemmatization
 iii) Stemming

OR

- Q6)** a) Given the confusion matrix, calculate accuracy, precision, Recall, Error rate with description on heart attack risk. [8]

		Predicted classes	
Classes		Heart-Attack Risk-yes	Heart Attack Risk-No
Actual Classes	Heart Attack Risk-yes	80	220
	Heart Attack Risk-No	150	9,500

- b) Explain the TF/IDF (term frequency-inverse document frequency) terms in text analysis with suitable example. [9]

Q7) a) List the data visualization tools and discuss any four applications of data visualization along with the use of the suitable plot. [9]

b) List the challenges of data visualization explain the types of visualization with example. [9]

OR

Q8) a) Explain in detail the Hadoop Ecosystem with suitable diagram [9]

b) Write a short note on the following [9]

i) Map reduce.

ii) Pig

iii) Hive



Total No. of Questions : 8]

SEAT No. :

P-3153

[Total No. of Pages : 2

[6003]-354

T.E. (Computer Engineering)

Data Science and Big Data Analytics
(2019 Pattern) (Semester - II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) What is Model Building elaborate this phase of data analytics with the help of suitable example. [9]

b) Explain any three sources of Big Data. Differentiate BI versus Data science. [8]

OR

Q2) a) What are the three characteristic of Big Data and what are the main consideration in processing Big Data. [8]

b) Explain Descriptive, Diagnostic, Predictive analytics. [9]

Q3) a) Explain why decision tree are used. Draw a sample decision tree and explain its parts. [9]

b) How Apriori Algorithm works, explain with suitable example? [9]

OR

Q4) a) What is data preprocessing? Explain in details about handling missing data and transformation of data. [9]

b) Explain Naïve Bayes' classifier and it applications. [9]

P.T.O.

- Q5)** a) What is text processing? Explain TF-IDF with example. [8]
b) With suitable example, explain the steps involved in k-means algorithm. [9]

OR

- Q6)** a) Define following terms with respect to confusion matrix : [8]
i) Accuracy
ii) Precision
iii) Recall
iv) AUC-ROC
b) Explain k-fold Cross Validation & Random Subsampling. [9]

- Q7)** a) With a suitable example, draw a Histogram, boxplot and explain its usages. [9]
b) Describe the data visualization tool Tableau. List of data visualization tools. [9]

OR

- Q8)** a) What is Data Visualization? Describe the challenges of data visualization. [9]
b) Explain architecture of Apache-Pig. [9]



Total No. of Questions : 8]

SEAT No. :

P-7545

[Total No. of Pages : 3

[6180]-53

T.E. (Computer Engineering)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6. Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is permitted.

- Q1)** a) Explain Data Analytics Cycle with suitable diagram and its phases. [8]
b) List and Explain the various activities involved in identifying potential data resources as a part of discovery phase in Data Analytics Life Cycle? [9]

OR

- Q2)** a) List and explain the key roles for successful analytics project. [8]
b) Write short note on : [9]
i) Common Tools for the Model Building
ii) Model selection for Data Analytics
- Q3)** a) List and explain the various types of analytics in Big data. [9]
b) Calculates the support and confidence value for all the possible item sets. [9]

Transaction ID	Items bought
1	Onion, Potato, Cold Drink
2	Onion, Burger, Cold Drink
3	Eggs, Onion, Cold Drink
4	Potato, Milk, Eggs
5	Potato, Burger, Cold Drink, Milk, Eggs

OR

P.T.O.

- Q4) a)** Explain the need of logistic regression along with its various types. [9]
- b)** Explain the following terms with suitable example. [9]
- Removing Duplicates from dataset.
 - Handling Missing Data

- Q5) a)** Suppose that the given data the task is to cluster points (with (x, y) representing location) into three clusters, where the points are A1 (2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1 and C1 as the center of each cluster, respectively. [8]

Use the k-means algorithm to show only show only the first round of execution with cluster center.

- b)** Explain the following Text Analysis steps with suitable example [9]
- Part-of-speech(POS)tagging
 - Lemmatization

OR

- Q6) a)** Given the confusion matrix, Calculate Accuracy, Precision, Recall, Error rate with description on Diabetic Risk. [8]

	Classes	Predicted classes	
		Diabetic Risk -Yes	Diabetic Risk -No
Actual classes	Diabetic Risk- Yes	90	210
	Diabetic Risk- No	140	9560

- b)** Explain the Text Preprocessing steps with suitable example. [9]

- Q7) a)** List the few data visualization tools and discuss any four applications of data visualization along with the use of the various plots with Python/R or suitable tool. [9]
- b)** List the challenges of Data Visualization. Explain the types of visualization with example. [9]

OR

- Q8)** a) Explain in detail the Hadoop Ecosystem with suitable diagram along with the various components. [9]
- b) Write a short note on the following. [9]
- a) Map Reduce
- b) Pig



Total No. of Questions : 8]

SEAT No. :

PB4430

[6262] 43

[Total No. of Pages : 3

T.E. (Computer Engineering)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester- II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) What is the data Preparation phase in Data Analytics Lifecycle. What is the Analytics Sandbox and ETLT process in this phase? **[8]**

b) List out different stakeholders of an analytics project. What they usually expect at the conclusion (key outputs) of a project? **[8]**

OR

Q2) a) List out the activities to be carried out in model planning and model building phase. What are different tools used for these phases? **[8]**

b) What is linear regression, and what are its primary objectives? What is the difference between simple linear regression and multiple linear regression? How do you evaluate the performance of linear regression? **[8]**

Q3) a) What is logistic regression, and how does it differ from linear regression? What is the sigmoid function, and what role does it play in logistic regression? **[9]**

b) Suppose you are given a dataset containing information about whether emails are spam or not spam, along with two features: the presence of the word "offer" (1 for present, 0 for absent) and the presence of the word "free" (1 for present, 0 for absent). You are tasked with classifying a new email with the following feature values: "offer"=1 and "free"=1. **[9]**

P.T.O.

Given the training dataset:

Email	Offer	Free	Spam
1	1	0	No
2	0	1	Yes
3	1	1	Yes
4	0	1	No
5	1	1	Yes

Calculate the probability that the new email is spam using Naive Bayes.

OR

Q4) a) How does the Apriori algorithm discover frequent itemsets in a dataset? What is the role of support and confidence in the context of association rule mining using the Apriori algorithm? [9]

b) Explain the process of building a decision tree? What are the criteria used for splitting nodes in a decision tree? [9]

Q5) a) Suppose you have the following dataset containing the coordinates of points in a 2-dimensional space: [9]

Point	X Coordinate	Y Coordinate
A	2	3
B	4	7
C	3	5
D	6	9
E	8	6
F	7	8

Perform K-means clustering on this dataset with $K = 2$. Assume the initial centroids to be (2,3) and (8,6). Compute the new centroids after each iteration until convergence, and assign points to their nearest centroids.

- b) How do you handle noise and irrelevant information in text data during preprocessing? Explain the terms bag of words and TF IDF in text analytics. [9]

OR

- Q6)** a) Explain how hierarchical clustering can be used for visualizing hierarchical relationships in data with suitable example? What are some real-world applications of hierarchical clustering? [9]

- b) What is the holdout method, and how does it work? Explain the difference between training set, validation set, and test set in the holdout method. [9]

- Q7)** a) What is a histogram? How is it used to visualize the distribution of data? How is it different from a density plot? [9]

- b) What is the Hadoop ecosystem, and what are its primary components? What is MapReduce, and how does it fit into the Hadoop ecosystem? [9]

OR

- Q8)** a) What is a box plot? Explain the different components of a box plot? How do you interpret the median, quartiles, and whiskers in a box plot? What does the interquartile range (IQR) represent in a box plot? [9]

- b) Explain the role of Apache Pig in data processing workflows on Hadoop? What is Apache Spark, and how does it complement Hadoop for big data processing? [9]



Total No. of Questions : 8]

SEAT No. :

PC1726

[6353]-43

[Total No. of Pages : 2

T.E. (Computer Engineering)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester-II) (310251)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks
- 3) Neat Diagram must be drawn wherever necessary.
- 4) Assume suitable data if necessary.

- Q1)** a) Draw data analytics life cycle diagram and briefly explain its phases. [6]
b) Explain the various key roles for a successful analytics project. [6]
c) What are various sources of Big data. [6]

OR

- Q2)** a) Describe few applications of Big Data Analytics. [6]
b) Explain Data preparation phase of data analytics lifecycle. [6]
c) List common tools used for model building phase of data analytic. [6]

- Q3)** a) Define and explain Entropy and Information gain. Calculate the entropy of the following distribution [9]

Fruit Color	Taste	Count
Yellow	Sweet	10
Red	Sweet	5
Green	sour	15
Orange	sour	5

- b) Explain Naïve Bayes Classifier. [8]

OR

P.T.O.

- Q4)** a) Explain Apriori algorithm with suitable example. [9]
b) Describe different categories of analytics [8]

- Q5)** a) What is Hierarchical clustering? Explain hierarchical clustering algorithms. [9]
b) Write a note on: [8]
i) Holdout method
ii) k-Fold Cross-Validation

OR

- Q6)** a) What is Text analysis? Explain the different steps involved in the text analysis. [9]
b) Write a note on Social network analysis. What are the applications of Social network analysis? [8]

- Q7)** a) Explain Hive architecture with suitable diagram. Describe characteristics and features of hive. [9]
b) Describe the data visualization tool Tableau. [9]

OR

- Q8)** a) What is data visualization and objectives of data visualization? Why it is difficult visualize Big Data? [9]
b) Write a note on Microsoft Power BI and Qlik [9]



Total No. of Questions : 8]

SEAT No. :

PD-5303

[Total No. of Pages : 2

[6403]-43

T.E. (Computer Engineering)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - VI) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is allowed.

Q1) a) What is Model Building elaborate this phase of data analytics with the help of a suitable example? **[8]**

b) List out different stakeholders of an analytics project. What do they usually expect at the conclusion (key outputs) of a project? **[9]**

OR

Q2) a) Explain Descriptive, Diagnostic, Predictive analytics. **[8]**

b) List and explain the various activities involved in identifying potential data resources as a part of discovery phase in Data Analytics Life Cycle? **[9]**

Q3) a) What is association rule mining? Describe the working of the Apriori algorithm with an example. **[9]**

b) Explain how decision trees are constructed using information gain and entropy. Illustrate with a small example. **[9]**

OR

Q4) a) Explain Naïve Bayes' classifier and its applications. **[9]**

b) Consider a dataset with binary classes and two features: "Loan Amount" and "Default History." Show how logistic regression could be applied for loan default prediction. **[9]**

P.T.O.

Q5) a) Explain the holdout method. Differentiate training set, validation set, and test set. [8]

b) Given the confusion matrix below. Calculate Accuracy, Precision, Recall and F1-score. [9]

	Predicted Yes	Predicted No
Actual Yes	70	30
Actual No	20	80

OR

Q6) a) Explain the following Text Analysis steps with suitable example [8]

i) Part-of-speech (POS) tagging

ii) Lemmatization

b) Use K-Means Clustering for the following points and determine the centroids after one iteration. Assume initial centroids as A(1,1), B(5,7). Points: (1,2), (2,1), (3,5), (6,8), (7,6), (5,5) [9]

Q7) a) Explain Hadoop Architecture with a neat diagram. Highlight the roles of NameNode and DataNode. [9]

b) Compare Tableau, Power BI, and Matplotlib for data visualization. Discuss scenarios where each tool is best suited. [9]

OR

Q8) a) What is Data Visualization? Describe the challenges of data visualization. [9]

b) Write short notes on the following : [9]

i) Map Reduce

ii) HDFS

iii) Hive

